JAVA PROJECT

Community Civic Issue Reporting System

CivicTrack

AJavaSwingandJDBCbased complaint management ap p l icat ion



PROJECT DETAILS

Student names

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Introduction

What is civic issue reporting?

- Citizens inform local authorities about problems in their neighbourhood
- Authorities track, prioritise and resolve each report

Why a digital system?

- Quick and simple complaint submission
- Every complaint is recorded and traceable
- Status is visible to citizens



Damaged roads

Potholes and broken surfaces



Garbage accumulation

Overflowingbins and litter



Drainage issues

Blocked drains, waterlogging



Streetlight problems

Faulty ormissing lights

Problem Statement



Time- c on su ming reporting

Traditional complaint submission takes effort and time



No proper tracking

Citizens cannot see how their complaint is pr o gr essi n g



Diffi c u lt prioritisation

Urgent issues are hard to identify among many



Limited communication

Weak link between citizens and authorities



U norga n ised records

Complaint data is scattered and hard to manage

The need

A centralised digital system for reporting, tracking and prioritising civic complaints

Proposed Solution: CivicTrack

CivicTrackdigitisesthecomplaintlifecycle:citizensreportandtrack, the system prioritises, and administrators resolve.



Report

Easy complaint submission through a Swing form

Track

Check status with a unique complaint ID

Prioritise

Automatic priority score and community upvotes

Administer

Dashboard to filter, update and manage complaints

Project Objectives

Simplify complaint submission	A user-friendly Swing form with all essential fields
Improve complaint tracking	Unique complaint IDs with current status
Help administrators prioritise	Automatic priority score highlights important issues
Encourage participation	“Me Too” upvotes show community concern
Maintain organised records	All complaints stored in a SQL database

Existing System vs Proposed System

Aspec t	Existing (traditional) system	Proposed system: CivicTrack
S u bmission	Time-consuming, manual complaint filing	Simple Swing form with required details
Tr a c kin g	No proper tracking of complaint pr o gr ess	Track any complaint using its unique ID
P r ior itisa tion	Urgent issues are hard to identify	Automatic priority score, updated by u pvo tes
Communication	Limited citizen–authority communication	Visible status and administrative r ema r ks
Records	Unorganised, scattered records	Structured SQL storage and admin dashboard

Technologies Used



Java

PROGRAMMING LANGUAGE

Core language for all application logic and OOP d esign



Java Swing

GUI FRAMEWORK

Builds the forms, tables and admin dashboard windows



JDB C

DATABASE CONNECTIVITY

Connects Java code to the database and runs SQL



SQL Database

RELATIONAL DATABASE

Stores complaint records and their status



IntelliJ IDEA

IDE

Used for coding, debugging and project management

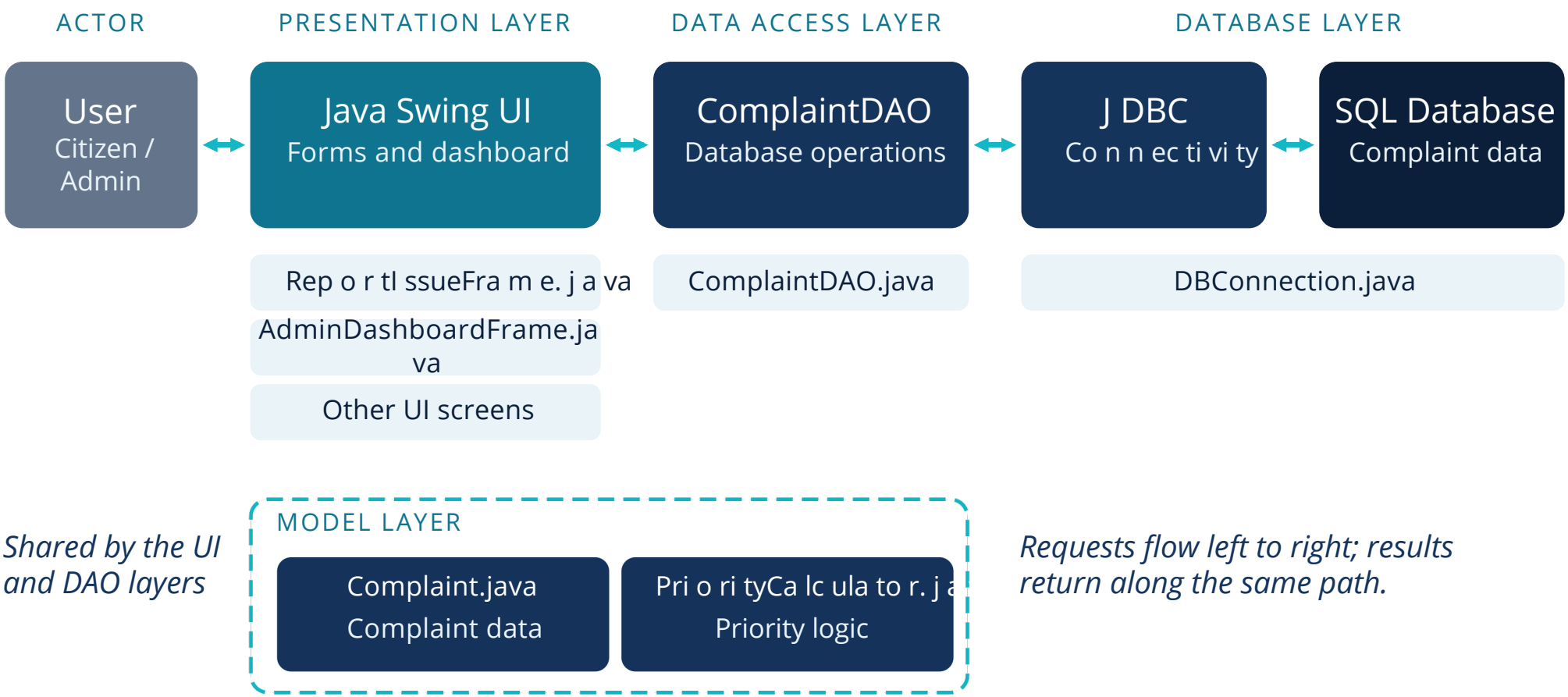


Maven

BUILD TOOL

Manages dependencies and automates project builds

System Architecture



Main System Modules

Citizen Module

- Submit complaints with full details
- Receive a unique complaint ID
- Track status using the ID
- View complaint details
- Upvote issues with “Me Too”

Administrator Module

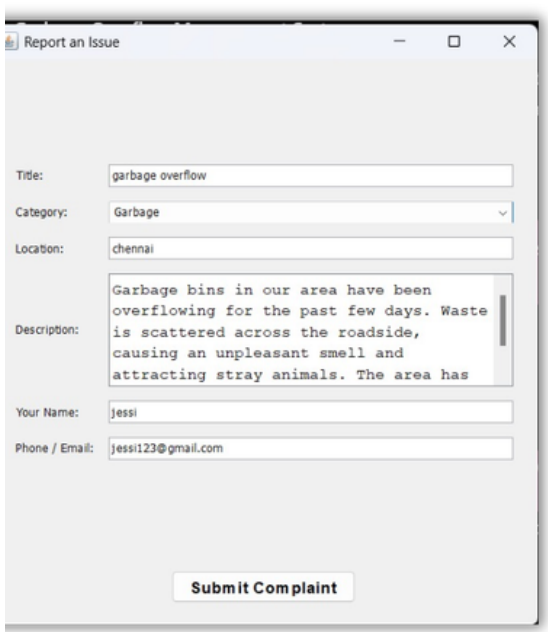
- View all complaints
- Filter complaints by status
- View complaint details
- Update complaint status
- Add administrative remarks
- Delete complaints
- Monitor dashboard statistics

Citizen Interface and Complaint Reporting

1. Enter details
2. Validate fields
3. Calculate pr io r i ty
4. Save complaint
5. Generate complaint ID

Complaint form fields

- Title
- C a tego r y
- Location
- Description
- Reporter name
- Contact information



Complaint Tracking and Upvoting

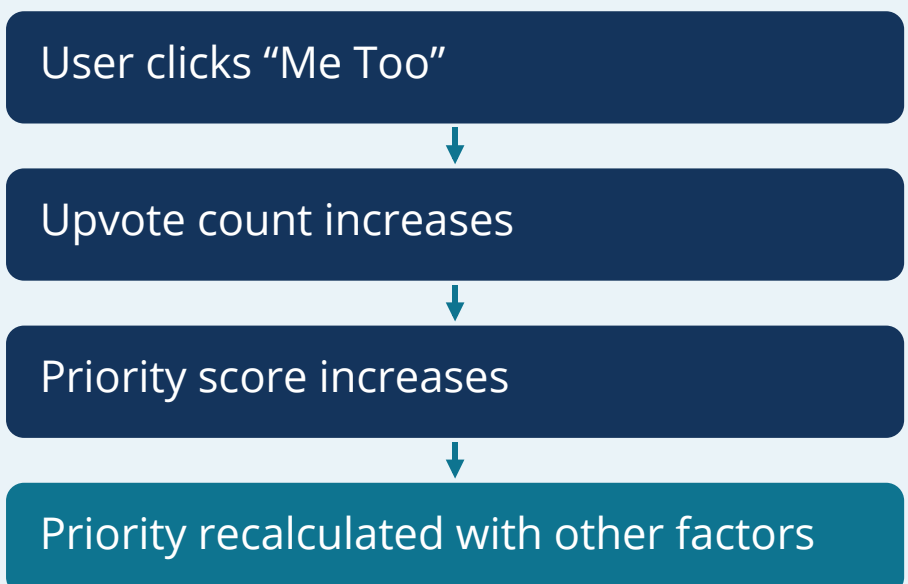
Complaint Tracking

- Every complaint receives a unique complaint ID
- Citizens enter the ID to view details
- Current status is shown with the details

Status values



Community Upvoting



Admin Dashboard

View all complaints

Complete listinone table

Filter by status

Pending, In Progress, Resolved

View details

Fullinformation of a com pl ai nt

Update status

Move complaints through the workflow

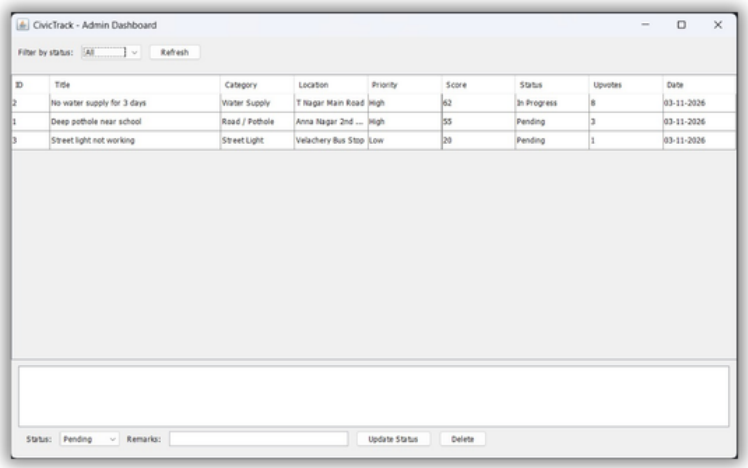
Add remarks

Administrative notes on a complaint

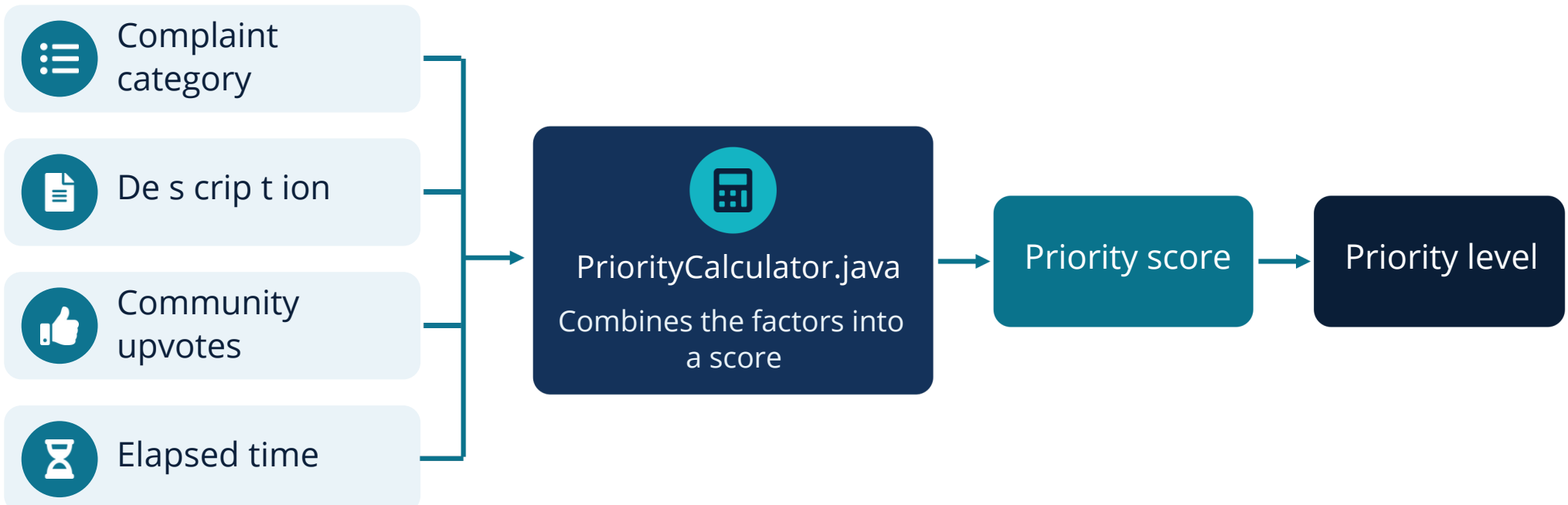
Delete complaints

Remove records when necessary

Dashboard statistics Pending In Progress Reso lved



Priority Calculation Mechanism



Factors contribute to the score; exact weights are defined in the source code

On submission: category and description set the initial score

On each upvote: priority is re cal cul ate d

Java OOP Concepts: Encapsulation and Inheritance

1. Encapsulation

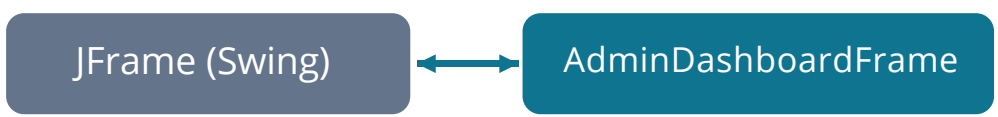
```
public class Complaint {
 private String title;
 public String getTitle() { return title; }
public void setTitle(String t) { title = t; }
}
```

Simplified example

- Complaint data is kept private inside the Complaint class
- Public getters and setters give controlled access
- Setters can enforce rules before data changes

2. Inheritance

```
public class AdminDashboardFrame
        extends JFrame { ... }
```



- AdminDashboardFrame is-a JFrame
- Inherits window features: title bar, size, visibility
- Promotes code reuse; no window built from scratch

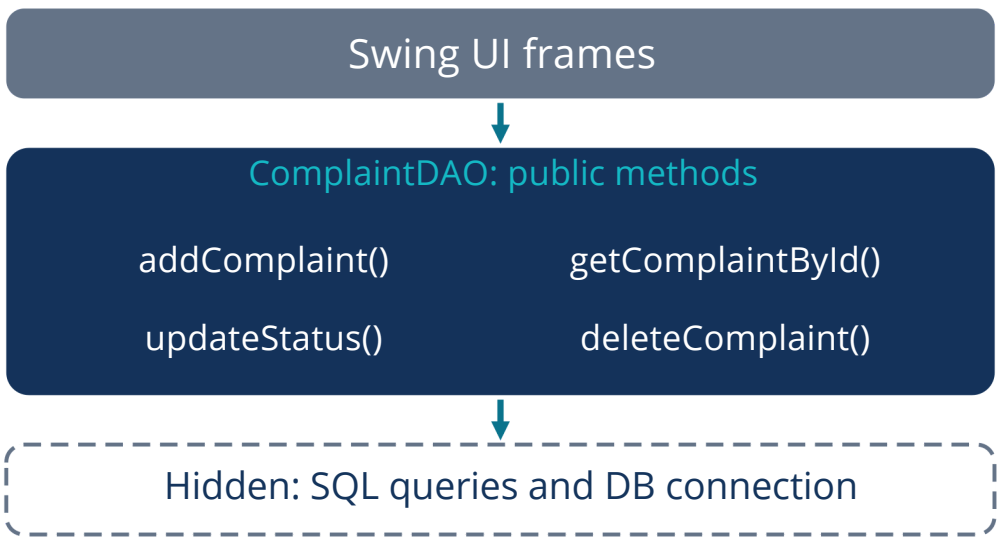
Java OOP Concepts: Polymorphism and Abstraction

3. Polymorphism

```
model = new DefaultTableModel(cols, 0) {
@Override
    public boolean isCellEditable(int r, int c) {
        return false;
    }
};
```

- Method overriding: isCellEditable() is redefined
  - Default behaviour (editable cells) is replaced by return false
  - Table data stays read-only
- Overridden method is chosen at runtime

4. Abstraction



- UI never writes SQL directly
- Database details can change without touching the UI

Database Management



Store

Saves new complaint details

I N SERT

Retrieve

Fetches records for tracking and dashboard

SEL ECT

U pd ate

Changes status and r ema r ks

UPDATE

Delete

Removes a complaint record

DELETE

Advantages and Benefits

Easy complaint submission
Simple, guided GUI form

Organised management
All complaints in one database

Improved transparency
Citizens can see current status

Better prioritisation
Scores highlight urgent issues

Community participation
Upvoting reflects public concern

Efficient admin workflow
Filter, update and remark in one place

Future Enhancements

Proposed features; not part of the current implementation

PROPOSED

Mobile application

Report issues on the go

PROPOSED

Image uploads

Attach photos to complaints

PROPOSED

GPS-based location

Auto-detect issue location

PROPOSED

Email / SMS alerts

Notify citizens of status changes

PROPOSED

Interactive maps

Viewcomplaintsona map

PROPOSED

Advanced analytics

Trends and reports for authorities

Conclusion

- CivicTrack digitises civic complaint reporting, tracking and resolution
- Automatic priority scoring and “Me Too” upvotes highlight urgent issues
- Layered Java design (Swing, DAO, JDBC) keeps the code organised
- The admin dashboard gives clear control over complaint status



Thank You

CivicTrack